Birmingham River Champion's
Invasive Species Overview

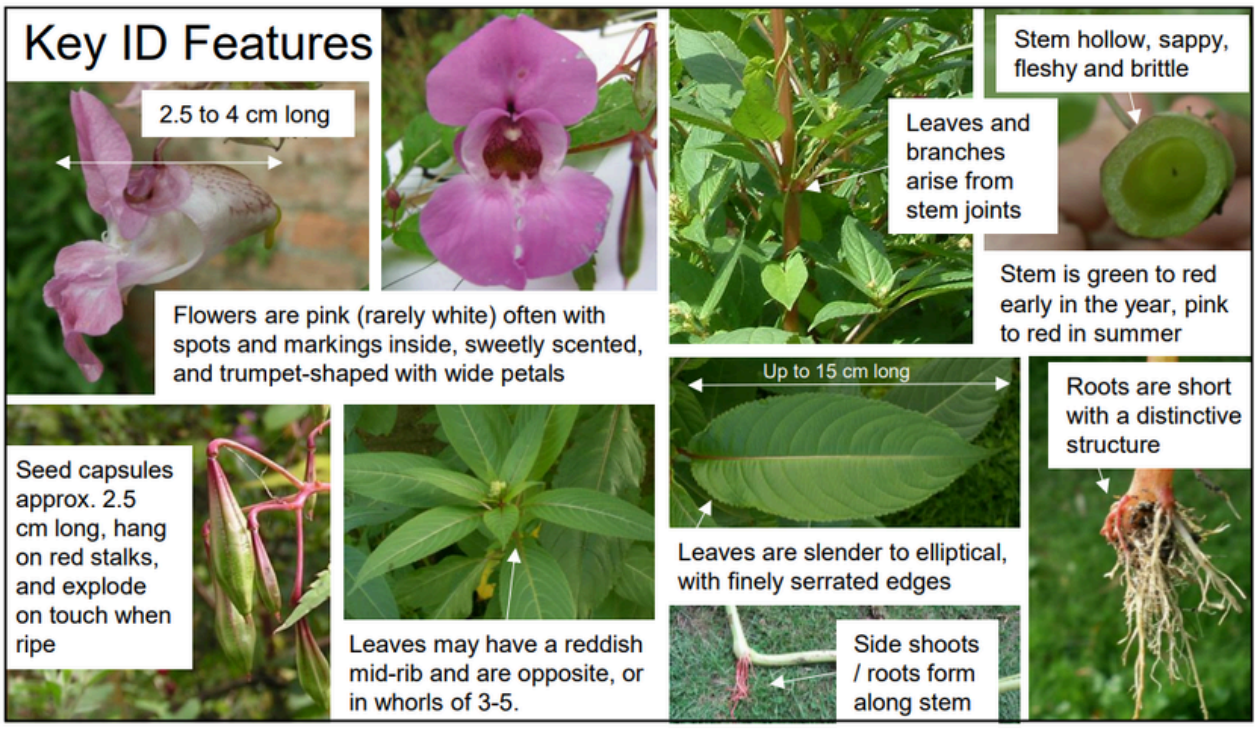
Invasive species, introduced accidentally or deliberately, can outcompete native wildlife, disrupt food chains, and damage habitats. Monitoring invasive species in rivers is essential to protecting the health of river ecosystems, allowing us to track their prevalence across Birmingham's watercourses (which could help target management interventions).

Volunteers can record the occurrence of invasive species either whilst carrying out other surveying (i.e., Urban Riverfly and/or water chemistry), or independently if you happen to see any whilst you are out and about.

For Birmingham River Champions, we have identified 5 invasive species to focus on that are prevalent regionally and nationally. This document provides information on each of these species, as well as images, from the relevant NNSS ID sheets.

Himalayan balsam (Impatiens glandulifera)

Native to the Himalayas and introduced to the UK in the 19th century as an ornamental garden flower, this plant is easily identified by its striking pink, helmet-shaped flowers that bloom from June-October. It can grow up to 2 meters and has long, serrated leaves with a reddish tinge. H. balsam often forms dense clusters along riverbanks, outcompeting native species, and its shallow roots contribute to soil erosion in winter.



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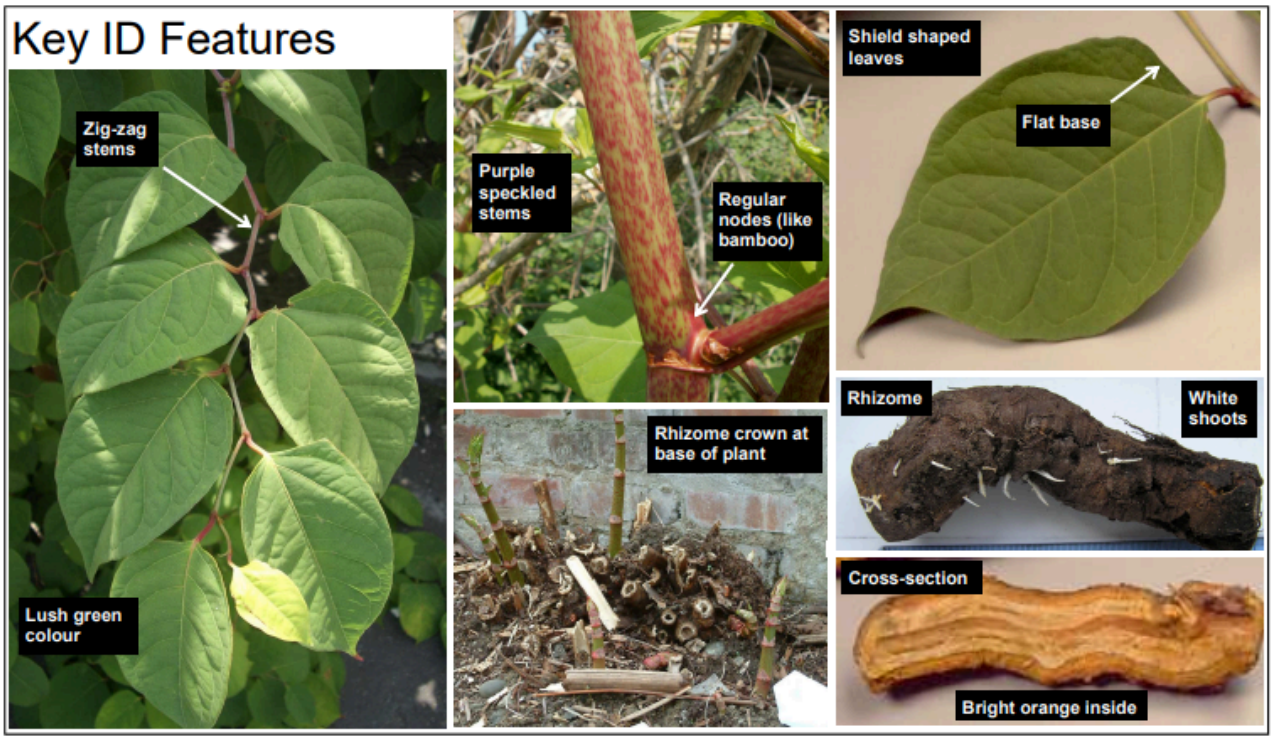
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Japanese knotweed (Fallopia japonica)

Originating from East Asia, this plant has bamboo-like stems, which are hollow and have distinctive purple speckles. Its heart-shaped green leaves are arranged in a zig-zag pattern along the stem, and produces small, creamy-white flowers in late summer.

It can grow up to 3 meters and spreads aggressively along riverbanks, damaging ecosystems and infrastructure with its dense root system.

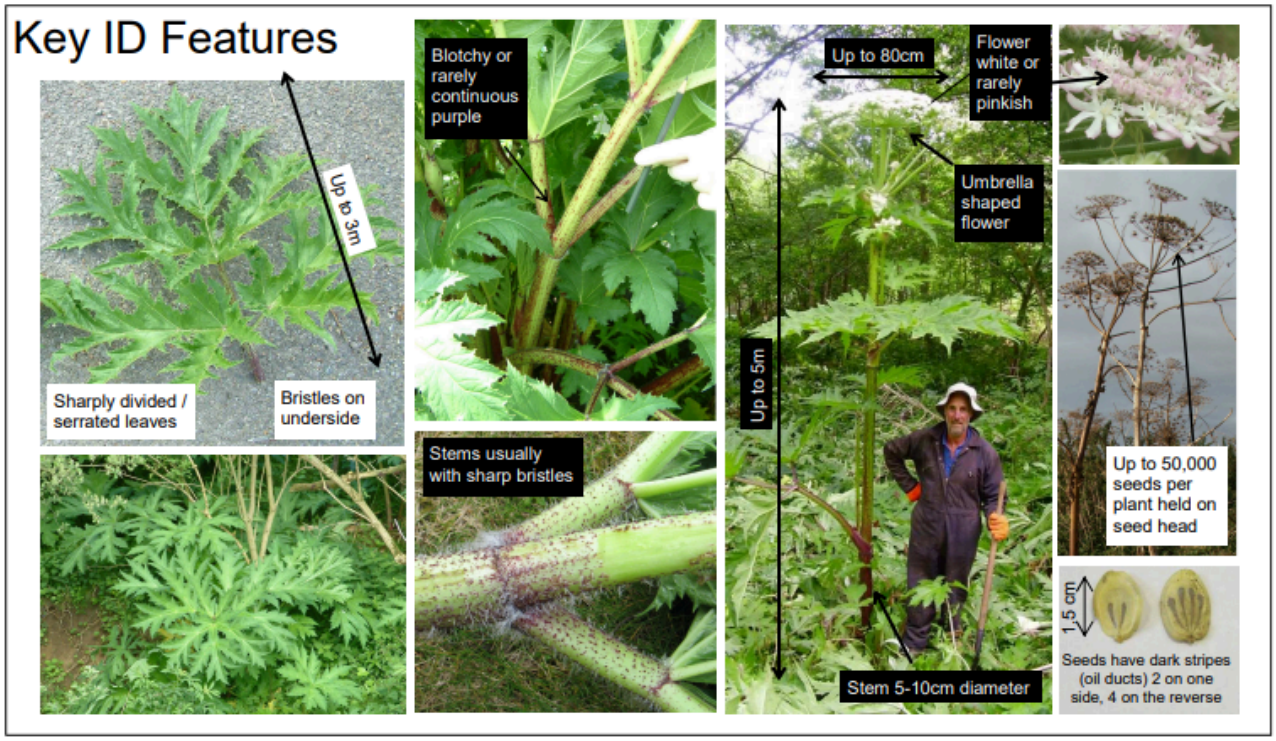


Giant hogweed (Heracleum mantegazzianum)

From the Caucasus region, giant hogweed is easily distinguished by its enormous size, reaching up to 5 meters tall. Its white, umbrellashaped flower heads can span up to 80 cm across, and it has large, sharply serrated leaves.

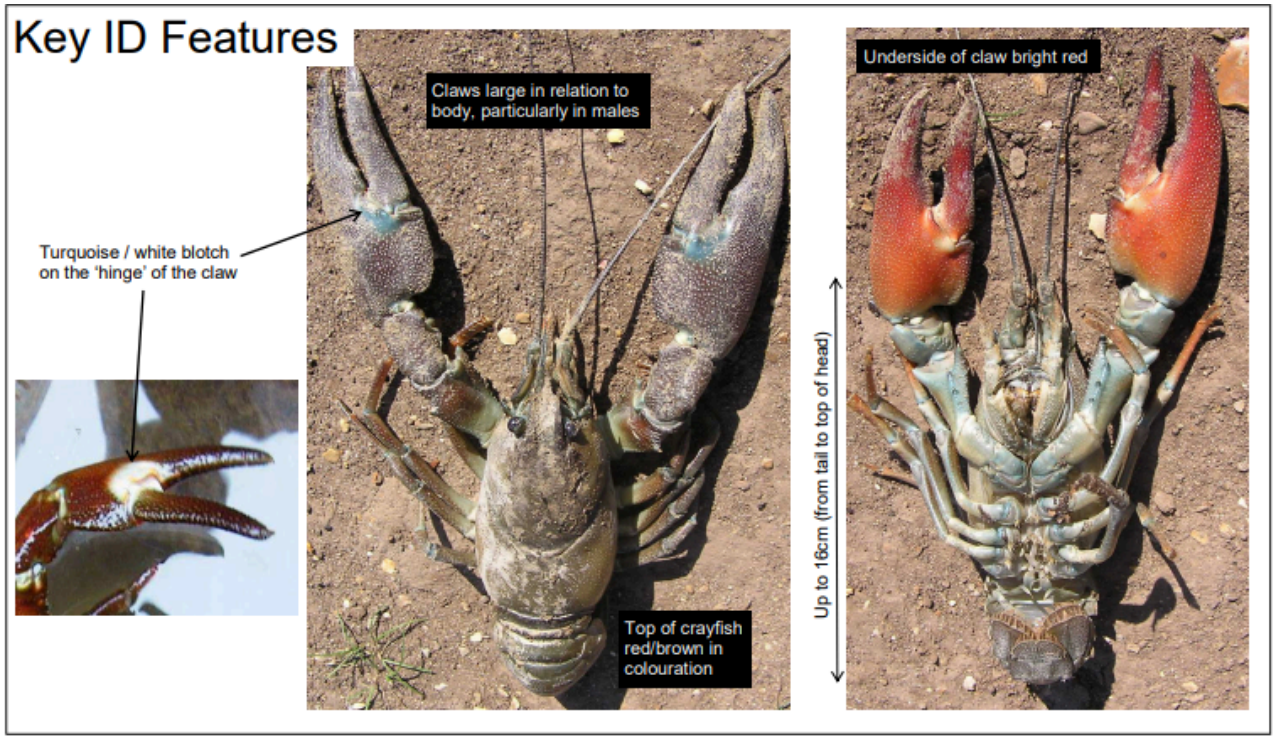
The stems are green with purple blotches and covered in coarse hairs. Most importantly, its sap is highly toxic and can cause severe burns to humans (see the risk assessment).

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Signal Crayfish (Pacifastacus leniusculus)

Introduced from North America in the 1970s, signal crayfish are significantly larger than the UK's native 'whiteclawed' crayfish, growing up to 16 cm long. They can therefore outcompete our natives, and further threaten their populations by carrying a fungal plague. Signals are identified by their large claws that are red on the underside and often have a distinctive blue or white patch near the joint (hence the name "signal"). Their burrowing activity destabilizes riverbanks.



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Killer/Demon Shrimp (Dikerogammarus villosus/ haemobaphes)

Originally from the Ponto-Caspian region, both the killer and demon shrimp are larger than native freshwater shrimp (Gammarus sp.), growing up to 30 mm in length. They often have a striped appearance, with spines on their tail and appendages, giving them a 'spiky' look – these features are not observed on our native shrimp.

Killer / demon shrimp are omnivores, and can outcompete native species for organic matter like leaf litter, or can predate on them directly.

Key ID Features

